

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

## Chapter 12: Surface Areas and Volumes

Previous year board questions, topic-wise: 3 questions

The tag after each question gives the board paper and the marks. Solutions are in a separate PDF with the same question numbers.

### 1 Surface area

1. A solid sphere is cut into two hemispheres. The ratio of the surface areas of sphere to that of two hemispheres taken together, is: (a) 1 : 1 (b) 1 : 4 (c) 2 : 3 (d) 3 : 2 **CBSE 2024, 1 mark**

### 2 Volume

2. The volume of the largest right circular cone that can be carved out from a solid cube of edge 2 cm is: (a)  $\frac{4\pi}{3}$  cu cm (b)  $\frac{5\pi}{3}$  cu cm (c)  $\frac{8\pi}{3}$  cu cm (d)  $\frac{2\pi}{3}$  cu cm **CBSE 2024, 1 mark**
3. The difference between the outer and inner radii of a hollow right circular cylinder of length 14 cm is 1 cm. If the volume of the metal used in making the cylinder is  $176 \text{ cm}^3$ , find the outer and inner radii of the cylinder. **CBSE 2024, 3 marks**